

Tool Improvement / Bug Fix Requests

1. Flood Tool — "Survey too large" error

Issue: When running the Flood tool's "From elevation" simulation on a full survey, the tool refuses to run with the message:

"The view is 1,878 m by 2,127 m — 734.1 million cells at this survey's 0.074 m resolution. This tool simulates at the survey's full resolution and never coarsens it, so it works over at most 12.0 million cells at a time. Draw a flood study area on the map — about 229 m square or less — and run it again."

Manually drawing a 229 m x 229 m box every time is impractical for large sites.

Requested fix / prompt to dev team: "The Flood simulation tool currently hard-caps at 12 million cells and forces the user to manually draw a small bounding box before it will run on large surveys. Please add one of the following so full-site flood simulation is usable:

- An option to run the simulation at a coarser/downsampled resolution automatically when the selected area exceeds the cell limit (with the resolution used clearly shown in the result), OR
- Automatic tiling: split the full survey into sub-tiles under the cell limit, run the simulation on each, and stitch/merge the results into one continuous flood layer, OR
- Move large-area runs to an asynchronous background job (with a progress indicator and notification when complete) instead of requiring real-time interactive computation. Please also let the user pick which mode (fast/coarse preview vs. full-resolution background run) to use."

2. On/Off Layer Control for Shapefile-Drawn Features

Issue: In the Shapefile tab, features (Point/Line/Polygon) can be drawn and exported, but there's no way to manage them as layers — they aren't grouped by type, can't be renamed, and can't be toggled on/off the way Ortho/DSM/DTM layers can.

Requested fix / prompt to dev team: "For features drawn in the Shapefile tool, please add proper layer management:

- Automatically group created features by geometry type (e.g., Polygons — Buildings/Hotels/etc., Polylines — Road lines/etc., Points — Locations/etc.).
- Allow the user to rename each individual created feature/layer from the dashboard.

- List these created layers in the same Layers panel used for Ortho, DSM, and DTM, with the same visibility toggle (eye icon / on-off switch) so users can show/hide their own drawn features alongside the base survey layers."
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3. KML Import Option (alongside Shapefile) with Independent On/Off Toggles

Issue: The "Upload, to compare" section under Shapefile only accepts a .zip shapefile. There's no KML import option, and once a file is uploaded there's no way to toggle its visibility — only remove it.

Requested fix / prompt to dev team: "Please extend the Shapefile upload panel (or add a parallel panel) to also accept KML file uploads for comparison, not just shapefiles (.zip).

- Both the uploaded Shapefile and the uploaded KML should appear as separate items in the Layers list.
 - Each uploaded file needs an on/off visibility toggle (like Ortho/DSM/DTM), so the user can hide/show it without deleting it.
 - Do not force the user to remove an existing uploaded shapefile before adding a KML, or vice versa — both should be able to coexist and be toggled independently."
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4. Cross Section / Cut & Fill Tools Fail on Large Surveys

Issue: On large surveys, both the Cross Section tool and the Cut & Fill tool return:

┆ "This survey is too large to measure interactively. Ask us for the figures you need."

This makes the tools effectively unusable for full-site analysis on larger datasets, forcing a manual support request every time.

Requested fix / prompt to dev team: "The Cross Section and Cut & Fill tools currently block entirely on large surveys instead of returning a result, showing 'This survey is too large to measure interactively.' Please resolve this by:

- Running the calculation as a background/async job for large surveys instead of requiring real-time interactive computation, with a progress indicator and the result delivered once ready (in-app notification and/or downloadable report), OR
 - Automatically computing at a reduced sample resolution for large areas (clearly labeled with the resolution used), with an option to request a full-resolution run separately. The goal is for users to get their cross-section heights and cut/fill volumes directly in the tool without needing to file a manual request for every large survey."
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5. CSV Download for Roads Tool — Sections / Chainage / Corridor Data

Issue: The Sections tool (under Roads/Tool) generates a chainage table with Centre/Low/High heights at set intervals, but this data is only viewable on-screen — there's no way to export it.

Requested fix / prompt to dev team: "Please add a 'Download CSV' button to the Roads tool's Sections (and Chainage/Corridor) output, similar to the existing 'Download Shapefile' button in the Shapefile tab. The exported CSV should include, per row, at minimum:

- Chainage (distance along alignment)
 - Centre elevation
 - Low elevation
 - High elevation
 - (Optional/nice-to-have) Left offset elevation, right offset elevation, cut/fill delta if a design level is set, and the sample interval/half-width used for that run. This should work for whichever interval the user selected (5 m/10 m/20 m/25 m) and for Corridor and Benches modes too, not just visualization on the map/table panel."
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6. Cross Section Tool — No On-Map Elevation Profile Display

Issue: When a cross-section line is drawn on the map, the tool only reports summary figures (Length, Points, Heights) in the side panel — it does not render an elevation profile chart of the line itself. Older desktop workflows show the section line drawn directly over the elevation surface with a linked profile graph underneath, updating live as the cursor moves along the path (location, elevation, and distance called out at the cursor). The web tool currently gives none of that visual/interactive feedback, only a static table of stats.

Requested fix / prompt to dev team: "Please add an on-map elevation profile view to the Cross Section tool, matching the interactivity of our legacy desktop tool:

- Once a section line is drawn, display a linked profile graph (distance along X-axis, elevation along Y-axis) below or alongside the map, generated from the same line.
- As the user hovers/drag along the drawn line on the map, sync a marker on the profile graph (and vice versa), showing live Location (coordinates), Elevation, and Distance-along-line at that point.
- Keep the existing Length/Points/Heights summary panel, but add the graph as the primary visual output so users can read the shape of the ground, not just the endpoint stats.
- This should degrade gracefully (e.g., coarser sampling) rather than disappearing entirely on large surveys, consistent with the fix requested in item 4."

7. No Option to Download Generated Basins as a Multipolygon

Issue: In the Water tab's Sink Detection tool, once "Basins" are generated (showing where each part of the survey drains to) and Depressions are found, the results are only viewable on screen — as shaded polygons on the map plus summary stats (Area, Storage, Deepest). There's no export option for these generated layers, unlike the Shapefile tab, which already has a "Download Shapefile" button for drawn features.

Requested fix / prompt to dev team: "Please add a download option for the Water tool's generated hydrology layers, starting with Basins:

- Add a 'Download' button (Shapefile and/or GeoJSON) next to the Basins layer, exporting it as a multipolygon — one feature per basin, not dissolved into a single combined geometry.
 - Include per-basin attributes in the export: Area, Storage, and Deepest point at minimum, plus the depression threshold used (e.g., "0.25 m or deeper") and the survey/date the hydrology was computed over.
 - Extend the same Download capability to the Channel network and Depressions layers in the same panel, using a consistent Download button that matches the existing pattern in the Shapefile tab, so hydrology outputs can be brought directly into GIS software."
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8. Cross Section Should Follow Active Layer(s) — DSM vs. DTM vs. Both

Issue: The Cross Section tool does not appear to distinguish which elevation layer is currently active when generating a section. Whether DSM, DTM, or both are turned on, the tool should reflect exactly what is toggled on, but this behaviour isn't currently in place.

Requested fix / prompt to dev team: "Please make the Cross Section tool aware of which elevation layer(s) are currently turned on, and generate the section accordingly:

- When only DTM is turned on, the cross section should show only the DTM (bare-earth) profile for that line.
- When only DSM is turned on, the cross section should show only the DSM (surface, including buildings/vegetation) profile for that line.
- When both DSM and DTM are turned on together, the cross section should plot both profiles on the same graph (e.g., two distinctly coloured/labelled lines) so the user can directly compare surface vs. ground elevation along the line, rather than only ever seeing one or a blended result.
- This should apply consistently whether the section is read from the summary panel or from the on-map elevation profile graph requested in item 6."